Geo-Yield-AI — AWS Migration Assessment

Production Architecture: Amazon RDS (PostgreSQL + PostGIS + pgvector), Amazon EC2 (FastAPI), & S3 + CloudFront (Vue 3)

Target Region:	**eu-central-1** (Frankfurt) / eu-west-1	**Date:**	October 2026
Database Stack:	RDS PostgreSQL 18 (PostGIS 3 + pgvector)	**Backend Compute:**	Amazon EC2 (t4g.small / Graviton3)
Frontend Delivery:	Amazon S3 + CloudFront (OAC / SPA)	**AI / Vector RAG:**	Amazon Bedrock (Titan V2 - 1024d)

1. Executive Summary & Strategic Objectives

Geo-Yield-AI combines spatial sociodemographic analytics with an artificial intelligence legal assistant (RAG) specialized in Barcelona municipal urban planning (Pla General Metropolità). The project currently operates locally via Docker Compose. This assessment formulates the definitive engineering strategy to migrate the application into a highly available, secure, and cost-efficient production environment in **Amazon Web Services (AWS)**.

The proposed architecture establishes a strict separation of concerns: persistent spatial and vector data are delegated to a managed **Amazon RDS for PostgreSQL** instance, the core FastAPI backend and Server-Sent Events (SSE) engine run on an optimized **Amazon EC2 Graviton3** instance, and the Vue 3 single-page application is delivered globally through **Amazon S3 and CloudFront** edge caching.

> ***Key Architectural Principle: Zero Plaintext Credentials.*** *By running the backend on EC2 with an attached **IAM Instance Profile**, the API invokes Amazon Bedrock (Titan Text Embeddings V2 & Reranker) and streams logs to Amazon CloudWatch through automatically rotated temporary AWS STS credentials, removing static API keys.*

2. End-to-End AWS Target Architecture

Layer	AWS Service & Component	Configuration & Specifications	Operational Responsibilities
Edge / CDN	Amazon CloudFront	HTTPS (port 443), TLS 1.3, Origin Access Control (OAC), SPA Fallback (403/404 -> /index.html 200 OK)	Global SSL termination, edge caching, static asset distribution, and optional /api/* proxying to EC2.
Frontend	Amazon S3 (Private Bucket)	Standard S3 Bucket with Block Public Access enabled; read policy granted strictly to CloudFront OAC Service Principal.	Hosts compiled Vue 3 / Vite assets (HTML, vanilla JS, CSS, SVG icons). Zero server maintenance.
Backend API	Amazon EC2 (ARM64 / Graviton3)	t4g.small (2 vCPU, 2 GB RAM). Docker container running deployment/Dockerfile or systemd. Nginx reverse proxy.	FastAPI orchestration, geocoding endpoints, PGM legal retriever, and SSE streaming (buffer disabled).
Database	Amazon RDS for PostgreSQL	PostgreSQL 18 (e.g. 18.6), db.t4g.small, 20-50 GB gp3 storage. Extensions: **postgis** and **vector** (pgvector).	Stores census data, commercial competitors, spatial layers, and 1024-dimensional legal chunk vectors.
AI Services	Amazon Bedrock	Embeddings: amazon.titan-embed-text-v2:0 (1024d) Rerank: amazon.rerank-v1:0 (eu-central-1)	Serverless inference invoked directly via Boto3 / SigV4 authentication without maintaining ML models on disk.

3. Deep Dive by Component

3.1 Relational & Spatial Database (Amazon RDS for PostgreSQL)

• **Engine & Extension Parity:** RDS PostgreSQL 18 supports both postgis and vector. Before restoring data, the database parameter group and initial DDL must execute: CREATE EXTENSION IF NOT EXISTS postgis; CREATE EXTENSION IF NOT EXISTS vector;.
• **Vector Space Upgrade:** In accordance with Alembic migration 0006_titan_embeddings_dim.py, the legal_chunks.embedding column is configured as vector(1024) with an HNSW cosine index (vector_cosine_ops).
• **Network Isolation:** RDS must be placed in a private DB Subnet Group across two availability zones with PubliclyAccessible=false. Inbound traffic on port 5432 is strictly permitted only from the EC2 security group.
• **Data Migration:** Logical dump via pg_dump --format=custom --no-owner --no-acl from the local Docker container, followed by pg_restore --verbose --exit-on-error against the RDS target as detailed in the runbook.

3.2 Backend API & Streaming Server (Amazon EC2)

• **Compute Form Factor:** An AWS Graviton3 t4g.small instance running Linux. Graviton processors deliver up to 20% better price/performance for Python FastAPIs and mathematical/GIS libraries compared to standard x86.
• **Reverse Proxy & SSE Compatibility:** Nginx (or Caddy) sits in front of FastAPI. Crucially, the endpoints /api/reports/stream and /api/chat/informe/stream transmit live token streams. Nginx must explicitly disable response buffering: proxy_buffering off; proxy_cache off; proxy_read_timeout 300s;.
• **IAM Instance Profile:** The EC2 instance assumes an IAM role with policies for Amazon Bedrock (bedrock:InvokeModel) and CloudWatch Logs (logs:PutLogEvents).

3.3 Frontend Application (Amazon S3 + CloudFront)

• **Static Distribution:** The Vue 3 application built with Vite generates pure static assets (HTML/JS/CSS). Amazon S3 acts as the object store without public website hosting enabled.
• **CloudFront with Origin Access Control (OAC):** Restricts S3 bucket access solely to the CloudFront distribution.
• **Client-Side Routing:** Vue Router HTML5 history mode requires that non-file paths return index.html. A CloudFront custom error response is configured: HTTP 403/404 -> /index.html with HTTP 200 status.
• **Unified Domain & CORS Elimination:** CloudFront can route /api/* to the EC2 backend and all other requests to S3. Both frontend and backend share the identical origin (domain), completely eliminating browser CORS errors.

4. Networking & Security Architecture

Security Group	Direction	Protocol / Port	Source / Destination	Purpose
sg-backend (EC2 Instance)	Inbound Inbound Outbound	TCP 80, 443 TCP 22 All traffic	0.0.0.0/0 (or CloudFront) Admin IP / SSM Session 0.0.0.0/0	Accepts web traffic and SSE streams. Admin access via SSH or SSM. Outbound to RDS, Bedrock, and Internet.
sg-database (RDS PostgreSQL)	Inbound Outbound	TCP 5432 None	**sg-backend ONLY** -	Allows SQL queries strictly from the backend API. Zero public exposure.

5. Sizing & Estimated Monthly Cost (eu-central-1)

Pricing based on official AWS EU (Frankfurt) standard commercial rates as of late 2026. Numbers reflect production MVP workload with moderate traffic.

AWS Resource	Dimension / Instance Class	Pricing Metric	Monthly Cost (EUR)
RDS PostgreSQL	db.t4g.small (2 vCPU, 2 GB RAM, Single-AZ)	~$0.0416/hr + 20 GB gp3 ($0.115/GB)	**~ €31.00**
EC2 Backend API	t4g.small (2 vCPU, 2 GB RAM, 20 GB gp3 root)	~$0.0168/hr + 20 GB gp3 ($0.08/GB)	**~ €14.50**
Elastic IP Address	1 Public IPv4 address assigned to EC2	$0.005/hr in-use IPv4 charge	**~ €3.40**
S3 Storage & Transfers	< 1 GB static assets, GET/PUT requests	Standard tier storage & API calls	**< €0.50**
CloudFront CDN	Edge caching, SSL termination, egress transfer	Within 1 TB/month Always Free Tier	**€0.00 (Free Tier)**

Bedrock & Observability	Titan Text Embeddings V2 + CloudWatch logs	$0.00002/1k tokens + log ingestion	**~ €3.00 – €5.00**
TOTAL ESTIMATED MONTHLY SPEND	**Full Production Stack**	**On-Demand Rates Baseline**	**~ €52.40 – €55.00 / month**

6. Step-by-Step Migration Roadmap & Cutover Plan

- **Phase 1: AWS Foundation & Networking**
  Create custom VPC with 2 public subnets (Internet Gateway) and 2 private subnets. Define sg-backend and sg-database security groups. Create IAM Instance Profile with bedrock:InvokeModel and CloudWatch write permissions.
- **Phase 2: Database Provisioning & Schema Restoration**
  Launch RDS PostgreSQL 18. Enable postgis and vector extensions. Export local database with pg_dump --format=custom --no-owner --no-acl -d geoyield -f geoyield.dump. Restore into RDS via pg_restore --verbose --exit-on-error. Validate Alembic revision matches 0006.
- **Phase 3: Backend EC2 Configuration**
  Launch EC2 t4g.small instance. Attach the IAM Instance Profile. Deploy the backend using Docker (deployment/Dockerfile). Inject production .env with RDS DATABASE_URL (enforcing sslmode=verify-full). Setup Nginx reverse proxy with SSE streaming.
- **Phase 4: Frontend Delivery (S3 + CloudFront)**
  Compile Vue 3 app with production API endpoint (npm run build). Create private S3 bucket and upload dist/. Create CloudFront distribution with Origin Access Control (OAC), ACM SSL certificate, and SPA error page routing.
- **Phase 5: Cutover & End-to-End Validation**
  Execute sanity checks on /health and /ready. Verify spatial endpoints (district boundary calculations, competitor radius queries). Verify Bedrock RAG vector search and streaming legal generation (/api/reports/stream).